

# TONY NGUYEN

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## EDUCATION

### UC San Diego

Master of Science in Computer Science, GPA: 3.73

La Jolla, CA

Graduated June 2026

### UC San Diego

Bachelor of Science in Computer Science, GPA: 3.94

La Jolla, CA

Graduated June 2025

## EXPERIENCE

### Software Engineer Intern

Feb 2024 – April 2025

Choir Power - AI/ML Energy Startup

La Jolla, CA

- Built a comprehensive React dashboard for B2B clients to visualize and analyze historical electricity usage patterns, detailed cost analysis metrics, ML-driven predictive consumption forecasts, and AI-Powered electricity usage tips
- Integrated AWS cloud services (Lambda, API Gateway, S3), implementing caching at the Lambda layer and optimizing API Gateway routing to reduce redundant data fetches, improving API response time by 30%
- Optimized performance with React code splitting, lazy loading chart libraries on-demand, and eliminating unnecessary re-renders through component memoization, reducing dashboard load time by 40% for 5 enterprise clients
- Participated in Agile development cycles including daily standups, bi-weekly sprint planning sessions, and collaborative code reviews

### Instructional Assistant of Software Engineering

Aug 2023 – Present

UCSD Computer Science Department

La Jolla, CA

- Design and lead weekly labs for Software Engineering course (200 students), creating labs and tutorials in Java
- Develop materials including practice problems and debugging guides used by 700+ students across multiple courses
- Conduct office hours for Software Engineering, Algorithms, Python Programming, and Theory of Computation
- Mentor student teams on software design patterns, OOP principles, and full-stack project development

## PROJECTS

### meap - Meal Prep Assistant | [Next.js](#), [React](#), [TypeScript](#), [SQL](#), [Supabase](#), [React Native](#)

Jan 2025 – Present

- Built comprehensive fullstack [nutrition tracking app](#) enabling users to create custom recipes, track daily macros, and manage ingredient/recipe inventory across web and mobile platforms
- Designed a dynamic ingredient and recipe backend with inventory checks and tracking, serving-based logging, and support for custom nutrition fields, ensuring accurate tracking and flexible food entry.
- Architected PostgreSQL database schema with user authentication, recipe approval workflow system, and real-time inventory management; designed REST API architecture for cross-platform use with full CRUD operations

### Toned - Workout Tracker | [Next.js](#), [React](#), [Javascript](#), [MongoDB](#)

Jan 2024 – Nov 2024

- Built fullstack [workout tracking app](#) enabling users to plan daily workout routines, log exercises with sets and reps, and visualize progress trends with interactive charts
- Designed a robust REST API with MongoDB empowering users with dynamic queries for seamless interaction with the database, including the ability to edit their workouts within the application using full CRUD operations.
- Implemented secure user authentication with AuthJS and built responsive, user-friendly UI optimized for mobile workout logging during gym sessions and desktop planning

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Java, Python, C/C++, SQL (PostgreSQL), HTML/CSS

**Frameworks & Libraries:** React, Next.js, React Native, Node.js, Express, Expo, Tailwind CSS, Flask, JUnit

**Tools & Technologies:** Git, GitHub, Supabase, MongoDB, AWS (Lambda, S3, API Gateway), Docker, Vercel

**Concepts:** REST APIs, Agile Development, Machine Learning Deployment, Authentication, Database Design, CI/CD

## RELEVANT COURSEWORK

Software Engineering, Object-Oriented Design, Advanced Data Structures, Machine Learning Algorithms, Machine Learning Optimization, Operating Systems, Mathematics for Algorithms and Systems, Design and Analysis of Algorithms